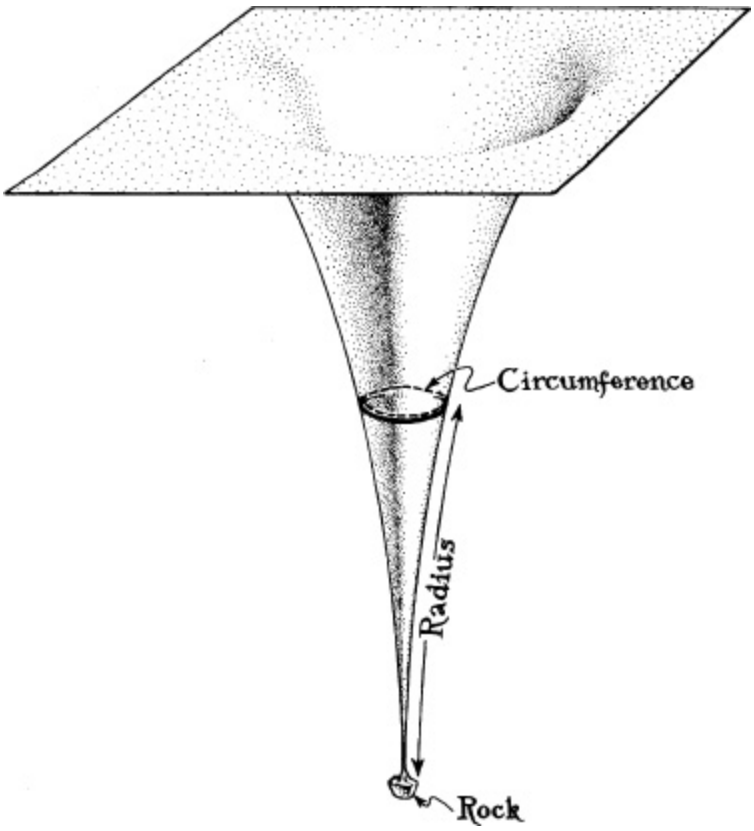
enormous. The horizon's radius is simply not computable from the meager information you possess: the hole's mass and its angular momentum.



P.3 A heavy rock placed on a rubber sheet (for example, a trampoline) distorts the sheet as shown. The sheet's distorted geometry is very similar to the distortions of the geometry of space around and inside a black hole. For example, the circumference of the thick black circle is far less than 2p times its radius, just as the circumference of the hole's horizon is far less than 2p times its radius. For further detail, see Chapters 3 and 13.

Abandoning your musings about the hole's interior, you prepare to explore the vicinity of its horizon. Not wanting to risk human life, you ask a rocket-endowed, 10-centimeter-tall robot named Arnold to do the exploration for you and transmit the results back to your starship. Arnold has simple instructions: He must first blast his rocket engines just enough to halt the circular motion that he has shared with the starship, and then he must turn his engines off and let the hole's gravity pull him directly downward. As he falls, Arnold must transmit a brilliant green laser beam back to the starship, and on the beam's electromagnetic oscillations he must encode information